

ActInf Textbook Group ~ Cohort 6 ~ Session 17 (Chapter 7, Part 2) 8/12/2024

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when um man that's
distracting
distracting okay I just won't share
screen it's what's causing
it where I was getting at was okay so on
the one hand we we have this
discretization and this continuity and
on the other hand uh we can we have this
embedding of the discreteness within the
continuity or the continuity uh within
the
discreteness um or there's like
nestedness right where you like you have
like discreteness continuity continuity
within the continuity and so on and so
forth but
also uh while we have the prioritization
of what's Happening there's also kind of
like a tuning aspect right like a tune
like there's tuning mechanisms
happening right like when you think of
stuff like hyperplanes and or hyper
parameterization um I think a beautiful
paper to look at um is biological
cognition by uh I think is hubner and
schuler and um they bring up this term
called
hierarchical processing right and
hierarchical has to do with like
selective tuning or you think of like a
like you're an audio engineer and you're
mixing you have a you have a mixing
board and you're trying to get the right
trying to mix the channels and all that
together the right way like this tuning

or this adaptive tuning rather but it's like there's like a mix of there's layers of the hierarchical and then n tuning happening right so so it's not even though I would argue it's mostly hierarchical you still have this tuning happening right you still have this sort of like this degrees of tuning happening but I'm not sure if that I like I'm I already am like about I'm on chapter five right now TR it's chapter four but I'm not sure if the book really addresses like this hyperparameterization issue but I guess you can kind of go over that if you can yeah yeah yeah that's um I think that's a really interesting point too so whenever I don't want to find the the figure I don't have it top of mine but it's I'm it's in the second half of the book and it might be in the section of learning but what are your what everything you're saying about like tuning and hyperparameters th those are absolutely like those are compon like directly components not just figuratively and in active inference models and so what what you can have is like like I mentioned Precision earlier that came up at the end of the the chapter this chapter um but you can actually like Precision can be modulated such that it impacts the rest of the model once you have a hierarchical model it's like it's not just you know every once in a while something happens here but most of the time it's over here it's like the whole

model is integrated right and so you do have this really strong constant interplay um between the different layers um so so that that that hopefully at least figure figuratively gives you some sense of like the the way you describe like a continuous and the discrete the discrete and the continuous it's like well as far as like coding a simulation you do have to more clearly Define one versus the other at least at each layer um but then that said they're so tightly integrated it's as if you know the discrete is impacting the The Continuous in given moment and vice versa so that's there and then somewhere in the in the in the textbook there's um there's another figure showing the PDP we were looking at earlier but it also adds on hyperparameters and those very hyperparameters are what can be adjusted or tuned and they relate to Precision uh for example you could have agent learn its a matrix meaning the the likelihood Matrix which figures in this chapter and so they might learn to you know let's say we we have our Mouse in the t- Maze the agent right let's say they repeatedly go through the maze over and over and through trial and error um end up learning that the stimulus is always on the left let's say we're we're the experimenters we to choose that let's say the stimulus is is always on the left for an agent who's who's who who who undertakes or engages in learning um it's not only going to try and minimize

its free

energy uh it's variational free energy

but it'll actually start to modify and

change its beliefs within that a matrix

to where it'll start to associate the

cheese or the attractive stimulus with

the location left in the Maze right it's

belief about that um and so that's

that's how that plays in but kind of

tweaking to the models and so on so we

have we have learning they H something

like hyper parameters here in the case

of

precisions U are directly implicated so

all that all that is here but that's a

really

interesting paper that you you brought

up um I almost feel like other folks uh

other members of The Institute might

find some interest in that um

found something heterarchical control

mechanisms in biology does that sound

right yeah

um if there's like any resources you

have to suggest for me to look into i'

would love to take a look at it because

I haven't really found something other

than that one paper I

mentioned it's biological bi cognition

um it's I think it's published through

Cambridge um

yeah it's the one through uh when you

look it up biological cognition it's the

Cambridge

article um it's paywalled but uh I was

able to get a copy of

it nice but uh yeah I I hate when when

like really nice papers are pay walled

it's like H it's such a crime you know
it's anyway yeah I mean I'm always
looking for more papers that talk about
this hyper mization stuff um because I
feel like there's just not enough out
there explaining that there's so much
stuff talking about hierarchy and like
embedding things in hierarchies and even
though I would say like the hyper
parameterization requires hierarchical
stuff in order to be effective it it's
still not

hierarchical in its nature but um yeah I
I you know definitely throw it at me I'm
I'm Catman do on Discord so if you want
to if you DM me that those resources if
you find them or even share it in one of
the channels I'll definitely I'm I'm
constantly eyeballing what's shared on
server so yeah in uh for the
relationship between discreet and
continuous in the machine learning
Paradigm what they actually use is a
mechanism called clip uh which is simply
going to take the energy relationship
between this categorical and this
discreet thing and then just throw it in
whatever's easier which happens to be uh
language in our case for uh large
language models so what you do is you
find the most straightforward
abstraction layer uh this is going to be
naturally hierarchical if we're talking
about deep learning in this specific
example uh you create that abstraction
layer and then you just keep it
consistent through the entire thing uh
two other things that might be related

to what you're talking about is
anthropics most recent paper uh which I
forgotten the name of I'll have to find
it but
they essentially talk about not
hyperparameters but the expectation of
the actual state that you're putting
into it which uh they Define as a
feature a feature is going to be deeply
related to what we consider to be a
concept like a dog or a cat or something
like that and being able to take these
Concepts which kind of create these
little bubbles that you can create a
context around there is some situation
that it's talking about which has some
set of features which are going to be
activated how much you activate these
certain features are going to be deeply
related to what you're saying about
these
hyperparameters so deciding which hyper
parameters to use is going to be found
right now in machine learning through
the creation of do we make this feature
really big do we shut these features
down to be really small and that's going
to be how we control how these
hierarchical mechanisms are actually
going
to consistently be able to produce the
results that we're looking for one other
slightly related thing is of the
application of going through the network
and having a kind of a meta Network
decide what inside of the network to
actually
create this is going to be a very recent

paper called uh it gets into a bit of the weeds but it's a mixture of a million experts which essentially takes the entire hierarchical network that we're going to be dealing with and all of the variables are going to be encoded into these one layer MLPs these one layer fully connected networks and from there we're going to find a way to create a meta Network which decides which ones to turn on and these ones that I'm talking about are going to be analogous to hyperparameters and it's taking a large Network which again talks about the context that you're speaking of and being able to decide which ones are important which facts are important inside of what we're talking about and only turning those on which I think is deeply analogous to the hyper thing but that does lead me to a question of whenever we're talking about hierarchies in the active inference domain how much does it matter you guys were talking about system one earlier how much does it matter with respect to continuous or discrete variables once you get past the system one level of the direct input from the external environment that you're trying to understand do you guys think that once you enter into a hierarchy it's all just abstracted away or do you guys think that the discreteness or the continuousness of the actual input is going to have really deep ramifications as you go down the line I'm sorry if that was a lot great

great comment we'll each just give a short closing thoughts in these last minutes I mean shortest answer it simply 100% dictates it it's like asking about a transistor how would it matter what the composition of the metal is or how would it matter what the size of the plate or how closely these elements are it's like it simply determines it so then there there might be some situations where you could discretize a continuous and get this or that performance retention or or the other way around but that's kind of the fun is the textbook is just showing us those fundamental motives and uh Toby SMI in the structured active inference live stream earlier today showed that in a in a far more comprehensive way than the Markov blanket formalism with the broader applied category theory for interfaces so it's like all about how it's specified like a physical transistor or Linux kernel deployment I also um this is this I don't know if this is a little too simple relative to to what we were all just discussing but I did drop a a link to one paper uh into the chat there and that is um deep active inference and scene construction and just given we have such little time left I just wanted to drop something there is U what with as a kind of combination of things we're talking about including hierarchical modeling um and here the idea of like looking at particular features that when combined maybe allows for let's say

classification or categorization of uh
in this case the paper I shared like a
visual stimuli it's rather interesting
they put together so a two-layer model
where um it's like a sod experiment so
it's like uh someone is viewing a 2 by
two grid and then they make inferences
based on particular things that they see
um in each of the um uh coordinates of
that 2 by two grid and then they uh from
there have to infer certain aspects of
what's going on in each of those two
grids and then that gets pushed up to a
higher discrete level that allows them
to uh to make an inference as to how
they should categorize the entire uh
scene the entire 2 by two grid like what
is going on in there it's based on a
previous experiment that's very similar
but uh it would just be um it's it's
a way to kind of
simplify uh a lot of the principles
we're sort of talking about here uh but
also being able to see something like
trying trying to theorize like how does
scene construction happen if you look at
broader uh expanse and you're trying to
figure out like infer a belief about
what is going on there well your eyes
are limited and so they can only look at
certain points so how do you start to
construct represent a belief about what
the overall scene is based on particular
features that you look at and you have
to make inferences about those
individual features in the first place
so you know for me to look at a you know
an overview of like a street and I don't

know exactly where you know that is in
the city I have to look at different
parts of the overall image I have to
infer oh that's a particular sign that's
particular building so on eventually to
the higher order inference of where am I
what what is this location so I just
thought that one would be interesting um
I'll try to add other things that that
maybe come to mind uh in the Cota um and
uh yeah really really interesting talk
today so thanks
everyone thank you fellows see you yeah
thank you both so much bye guys bye